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Reinforcement Learning for Adaptive Traffic Signal Control: An Overview

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Abstract

Traffic signal control is central to urban mobility because it directly influences congestion, travel time, and emissions. Traditional methods, including fixed-time signal scheduling and actuated control strategies, can handle normal conditions but often fail when traffic changes suddenly. Reinforcement learning (RL) has gained attention as a data-driven approach which adapts policies by interacting with traffic environments. Recent studies have investigated both single-agent and multi-agent formulations, encompassing value-based, policy-based, and actor–critic learning paradigms. This paper reviews how RL has been adopted for traffic signal control, grouping the core approaches and highlighting their evaluation practices. The review shows that most studies still focus on efficiency measures such as delay and queue length, with safety and environmental factors less frequently addressed. Nearly all evaluations are done in simulation, with SUMO and CityFlow as the dominant platforms. Key challenges remain in handling multiple objectives, scaling to large networks, and addressing the challenge of transferring simulation-based results to real-world deployment. By outlining current methods, their strengths and weaknesses, and the gaps that persist, this review points to the directions needed for RL to move from research to practice.

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1. Introduction

Managing traffic signals is a long-standing challenge for urban mobility. Early systems were based on fixed schedules that used historical traffic data [1]. These systems are simple but fail to adjust when traffic changes suddenly, for example during accidents or special events [2]. Later, actuated systems were introduced that use sensors to detect ve-

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hicles and adapt signals locally. Although more flexible than fixed-time control, actuated systems often remain limited to single intersections and cannot coordinate well across larger road networks [3].

In recent years, machine learning and artificial intelligence have opened new ways of handling signal control [4].Among these approaches, reinforcement learning (RL) has received significant attention because it enables agents to learn control policies from interaction data and adjust actions dynamically [5]. Its appeal for traffic lies in the ability to improve flow by reducing waiting times, lowering congestion, and even cutting emissions [6].

Early applications of RL in traffic signal control mainly used single-agent designs at isolated intersections. Over time, more advanced methods such as deep reinforcement learning (DRL) and multi-agent reinforcement learning (MARL) are being studied to coordinate multi-intersections and manage larger traffic networks [7]. These approaches allow richer state definitions, better reward functions, and cooperative decision-making strategies [8]. Several reviews have already studied this field. Some works focus on deep RL and its architectures [9], others compare algorithm families and computational aspects [10], while a more recent survey [11] examined recent deep reinforcement learning models for traffic signal control and discussed remaining research challenges.

In this paper, the focus is kept narrow: to give a short account of how reinforcement learning has been applied to traffic signal control, what types of methods are most common, and what results have been reported. Unlike existing surveys that focus primarily on specific deep reinforcement learning architectures or algorithmic implementations, this overview takes a broader perspective on how reinforcement learning has been evaluated and applied in practice. It compares algorithm families and discusses common choices in reward design, agent structures, and evaluation settings, with an emphasis on recurring trends and open limitations rather than on proposing or benchmarking new algorithms.

The main contributions of this overview are threefold: (1) it groups RL-based traffic signal control methods into value-based, policy-based, and actor-critic families; (2) it compares their strengths and weaknesses with respect to scalability, reward design, and deployment readiness; and (3) it identifies persistent gaps and outlines directions for future work. In doing so, the paper provides researchers and practitioners with a clear picture of where RL has been successful and where major challenges remain. Relevant studies were collected through a structured search of Scopus and Google Scholar, using keywords related to reinforcement learning and traffic signal control, including adaptive and multi-agent formulations. Most reviewed studies were published between 2022 and 2025. Earlier studies are included selectively to provide background and historical context, reflecting the progression of reinforcement learning–based traffic signal control rather than an exhaustive historical survey.

2. Background and Overview of RL Approaches

Traffic signal control strategies can be divided into three main groups: fixed-time, actuated, and adaptive. Fixed-time methods use pre-set schedules that repeat cycles based on past traffic demand. The work in [12] introduced one of the earliest formal descriptions of fixed-cycle signals, while another [13] provided a comprehensive review of traffic control strategies covering fixed-time and highlighting their respective strengths and limitations. Actuated methods respond to sensor input such as loop detectors or cameras. They are more responsive than fixed-time, but they usually operate only at the local level without coordination between intersections [14]. Adaptive traffic signal control (ATSC) builds on these ideas and uses algorithms to update signal timings in real time. Adaptive systems can be model-based, using predictive traffic models, or model-free, where the control strategy is learned directly from data.

Traditional adaptive systems such as the Split Cycle Offset Optimization Technique (SCOOT) and the Sydney Coordinated Adaptive Traffic System (SCATS) represent well-known model-based approaches, which rely on predictive traffic models and online calibration. The authors in [15] documented their global practices and noted that their effectiveness is strongly dependent on accurate modeling and frequent recalibration. These systems often struggle in environments with irregular or highly dynamic flows, where assumptions about driver behavior and demand no longer hold. Reinforcement learning offers an alternative by removing the dependence on fixed models and instead learning policies directly from observed states and actions. This ability to adapt in real time without requiring explicit traffic equations has become a key reason for the rise of RL in signal control research.

Different types of RL algorithms have been applied to traffic signals. Value-based methods, such as Q-learning and deep Q-networks (DQN), estimate the value of actions and select the one with the highest reward. They are effective for discrete tasks like phase selection but can be less scalable in large networks. Policy-based methods optimize the

control policy directly without relying on value estimation. Approaches such as Proximal Policy Optimization (PPO) are well suited to continuous action spaces, for example fine-tuning green times. Actor-critic methods combine both strategies, with one part of the model proposing actions (actor) and the other part evaluating them (critic). Examples include Deep Deterministic Policy Gradient (DDPG) [16], Soft Actor–Critic (SAC) [17], and Advantage Actor–Critic (A2C) [18], which are commonly used for large and complex urban networks. Hybrid approaches have also appeared, where RL is combined with other techniques like graph attention networks, fuzzy logic, or meta-learning. These frameworks aim to improve coordination across many intersections and adapt to diverse traffic situations.

Simulation tools are critical for testing these methods before real-world use. SUMO [19] is the most common platform, valued for its flexibility and open-source nature. In [20], they noted that SUMO dominates experimental studies, while the authors in [21] observed that CityFlow is increasingly adopted for large-scale experiments because of its speed and ability to handle thousands of intersections efficiently. A few studies have also used custom-built simulators. The choice of platform often depends on the required scale and realism of the experiments.

2.1. Reinforcement Learning

Reinforcement learning (RL) provides a way to improve decision making by learning from interaction with an environment. In the context of adaptive traffic control, the agent observes the traffic state, applies an action such as extending or switching a signal phase, and receives a reward that reflects traffic conditions. This process is commonly expressed as a Markov Decision Process (MDP) with states S, actions A, rewards R, and a discount factor γ [22]. Figure 1 illustrates a simplified reinforcement learning framework for traffic signal control at a signalized intersection, showing the interaction between states, actions, and rewards.

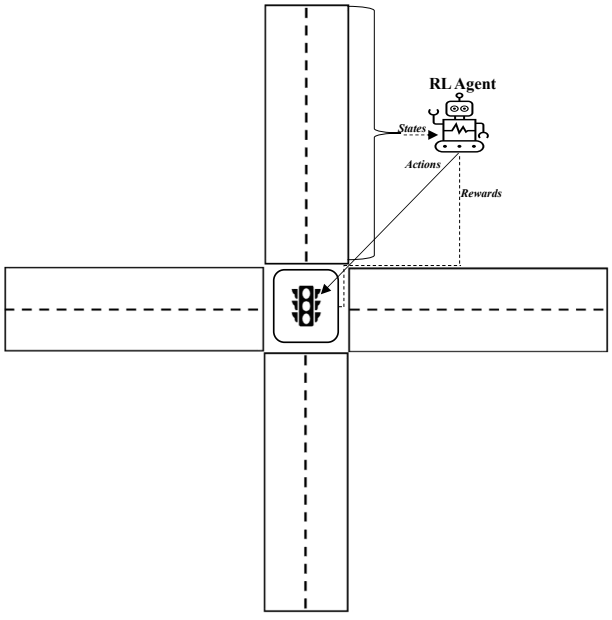


Fig. 1. A simplified reinforcement learning framework at a four-way intersection. The RL agent receives **state** information (e.g., queue length, density, waiting times), selects an **action** (e.g., change or extend a signal phase), and obtains a **reward** (e.g., reduced delay, improved throughput, lower emissions) from the traffic environment.

The return G_t is the total discounted reward from time t onward [22]:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}. \tag{1}$$

The state-value function under a policy π is defined as [22]:

$$v_\pi(s) = \mathbb{E}_\pi\Big[G_t \mid S_t = s\Big], \tag{2}$$

and the action-value function is defined as [22]:

$$q_\pi(s,a) = \mathbb{E}_\pi\Big[G_t \mid S_t = s, A_t = a\Big]. \tag{3}$$

Value-based methods focus on estimating $q_\pi(s,a)$, policy-based methods learn the policy $\pi(a|s)$ directly, and actor-critic methods combine the two [10].

A comparative overview of these RL families, including their strengths, limitations, typical applications, and representative implementations, is presented in Table 1.

Table 1. Comparison of Reinforcement Learning Approaches Used in Adaptive Traffic Signal Control Studies

Approach	Strengths	Limitations	Applications	Examples
Value-based (Q-learning, DQN)	Simple to implement; effective for discrete phase control	Limited scalability in large networks	Discrete phase selection at isolated intersections	[5, 23, 2]
Policy-based (e.g., PPO)	Handles continuous action spaces; stable in complex settings	Higher computational cost; sensitive to hyperparameters	Signal timing adjustment (e.g., green splits); traffic flow optimization in complex or corridor settings	[9, 24, 25, 26]
Actor–Critic (DDPG, SAC, A2C)	Balances discrete and continuous control	Training complexity; high computation	Adaptive control in complex networks	[27, 28, 29]
Multi-Agent RL (MARL)	Distributed; reflects real traffic networks	Coordination overhead; credit assignment issues	Network-wide coordination; city-scale systems	[30, 31, 7, 32, 33]

Table 1 highlights clear trade-offs across the main reinforcement learning categories used for adaptive traffic signal control. Value-based methods offer simplicity and effectiveness for discrete phase selection but face scalability limitations in large networks.Policy-based and actor–critic methods are well suited for continuous and complex control tasks, but they typically require greater computational resources and longer training times. Multi-agent reinforcement learning reflects the distributed nature of urban traffic networks and enables network-wide coordination, but introduces challenges related to communication overhead and credit assignment. These trade-offs explain the continued coexistence of different RL families in the literature, with no single approach uniformly dominating across all traffic scenarios.

3. Evaluation Insights

Research on RL-based traffic signal control is often assessed across a set of recurring dimensions. These include the design of the reward function, the structure of agents, the choice of performance indicators, and the simulation tools used. Together, they reveal how progress in the field is measured and compared.

3.1. Reward Functions

In RL, the reward function is central: it shapes agent behavior, steers learning toward desired objectives, and has major impact on how fast and how well the agent learns [34]. Most studies rely on reward functions that capture efficiency, typically through queue length, waiting time, or delay reduction [23]. Pressure-based designs, which balance inflows and outflows, have also been introduced to improve stability under congestion [35]. More recent work incorporates broader objectives such as safety and emissions [29], though these remain less common. A further challenge

is that combining objectives into a single function often requires subjective weighting. For instance, prioritizing delay over emissions or safety depends on design choices that may not reflect wider social goals. Developing transparent and principled ways of balancing such trade-offs is therefore a necessary step before real-world deployment. Hybrid designs that combine several indicators are emerging, showing a gradual move toward multi-objective optimization [36]. However, progress in this direction is still limited because multi-objective RL increases computational complexity and lacks standard benchmarks, making it harder to compare results across studies.

3.2. Agent Structures

Early applications of RL in traffic signal control were limited to single intersections with single agents. Recent work has shifted toward multi-agent reinforcement learning (MARL), where each intersection is modeled as an agent. More than half of the reviewed studies now use MARL, often supported by decentralized or graph-based coordination mechanisms [7, 36, 32, 30]. MARL better reflects the distributed nature of urban networks, though scalability and real-world deployment remain open issues. A central difficulty in MARL is the credit assignment problem: when several agents act simultaneously, it becomes difficult to determine which decision contributed most to the observed outcome. Communication overhead between agents also increases with network size, raising questions about how much coordination is needed and how it should be organized. These issues suggest that while MARL scales in principle, practical implementations will need to find the right balance between local autonomy and global cooperation.

3.3. Performance Indicators

The most common performance measures are delay and travel time, reflecting the primary aim of improving mobility. Several studies report significant reductions, such as [31] with 23.5% less delay for emergency vehicles. Queue length reduction is also frequently reported [23, 37].

3.4. Simulation Tools

Simulation environments remain essential for training and evaluation. SUMO is the dominant platform due to its open-source flexibility and wide adoption [20].Alternatives include VISSIM [38], Aimsun [39], and CityFlow [40]. CityFlow is increasingly used for large-scale scenarios because of its speed and ability to handle thousands of intersections efficiently [21]. A smaller number of studies use Paramics or custom-built tools. The choice of simulator often reflects the trade-off between scalability and realism. Equally important is the issue of reproducibility. Open platforms such as SUMO and CityFlow allow other researchers to replicate experiments, compare algorithms, and validate results, which helps the field move forward. In contrast, reliance on commercial simulators such as VISSIM and Aimsun can be a barrier, as they make it harder to share models openly and reproduce results. Establishing common experimental testbeds remains an open need.

4. Trends and Challenges

From the evaluation of recent studies, several recurring trends can be observed. At the same time, these trends reveal important limitations that still prevent large-scale deployment.

One clear trend is the continued prevalence of value-based approaches, with DQN and its variants being most commonly adopted for traffic signal control under discrete action settings. These approaches have proven effective in capturing traffic phase decisions, but their scalability to large networks remains limited and real-world deployment has not yet been achieved. Policy-based and actor-critic methods are gradually gaining ground, offering more stability in continuous or complex settings, though at higher computational cost. Early work focused on minimizing delays or queue lengths [41, 8], and while these approaches reduced delays and improved throughput, their objectives remained narrowly efficiency-driven.

Recent research has tried to move beyond efficiency by adding sustainability objectives. For instance, the work in [27] proposed a dual-objective algorithm that reduces both travel delay and emissions, while [28] developed an adaptive control method that balances CO_2 emissions with waiting times. Safety is another underexplored dimension.

Another work in [42] proposed SafeLight, which incorporates safety objectives into RL-based signal control, while others in [43] introduced a constrained method that balances efficiency and public transport priority with safety. Other studies have noted that when risk is not explicitly modeled, efficiency-driven RL policies may unintentionally encourage unsafe maneuvers [11].

A second trend is the widespread adoption of multi-agent reinforcement learning (MARL). More than half of the surveyed studies employ MARL, with intersections modeled as agents and decentralized or graph-based coordination schemes ensuring scalability. This reflects the distributed nature of real networks. With the rise of deep reinforcement learning (DRL) and MARL, the focus expanded to larger and more complex networks [33, 44, 45, 46]. For example, in [47] they combined max-pressure control with RL in mixed traffic, while the work in [48] applied multi-agent coordination with spatio-temporal hypergraphs at city scale. However, despite strong performance in simulation, MARL systems have not been transferred to real traffic environments, where communication delays, unpredictable driver behavior, and coordination complexity remain barriers.

Reward function design also shows a clear pattern. The majority of studies continue to use efficiency-based metrics such as delay and queue reduction. Pressure-based formulations improve balance under congestion, while safety- and emission-oriented objectives appear in only a few works. Despite these contributions, many models still rely on weighted sums of objectives, which is not a principled multi-objective approach. This gap underlines the difficulty of aligning RL optimization with broader sustainability and safety goals.

Another trend is the reliance on simulation as the testbed for RL systems. SUMO is the most widely used microscopic simulator and provides emission models for pollution analysis, but it scales poorly to very large networks. CityFlow, developed as a faster alternative, runs about twenty times quicker and is designed for MARL studies. VISSIM and Aimsun are used for high-fidelity modeling, but the diversity of platforms makes cross-comparison difficult. More importantly, the gap between simulation and deployment persists, as most results remain restricted to synthetic or controlled data.

Finally, the rise of hybrid and domain-specific methods highlights the trend toward combining RL with other techniques, such as graph networks, fuzzy logic, or meta-learning. These enhance adaptability but often increase computational requirements, which can conflict with the real-time demands of traffic control. Ensuring efficient and explainable models remains a challenge for practical deployment.

5. Conclusion

Reinforcement learning has become one of the most studied approaches for adaptive traffic signal control. Research has progressed from early single-agent designs at isolated intersections to multi-agent systems capable of coordinating hundreds or even thousands of signals in simulation. Value-based methods, particularly DQN and its extensions, remain dominant, while policy-based and actor-critic frameworks have expanded the algorithmic diversity. Evaluation practices continue to rely heavily on efficiency-oriented metrics such as delay and queue reduction, though recent work has started to integrate broader objectives, including emissions and safety.

Despite these advances, significant challenges remain. Most studies are still limited to simulation environments, with no large-scale deployment in real networks. Reward functions are often narrow in scope, computational demands of advanced models restrict real-time use, and the gap between simulation and reality persists. In addition, differences in traffic scenarios, simulation platforms, and evaluation metrics make reported performance gains difficult to compare across studies, highlighting the need for future systematic reviews or meta-analyses based on standardized benchmarks.

As connected vehicles and roadside sensors become more common, RL has the potential to act as the control layer that translates real-time data into adaptive decisions. Aligning optimization with broader goals such as reduced congestion, improved safety, and lower emissions will be essential for future deployment. This review has organized RL-based traffic signal control methods into value-based, policy-based, and actor-critic families, compared their strengths and weaknesses, and highlighted persistent gaps such as scalability, reward design, and the transfer from simulation to practice. By doing so, it provides both researchers and practitioners with a clear picture of the state of the field and the directions where further work is most needed.

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